

Experiment 7: simple substitution algorithm.

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char cipher[] =
```

```
    "53†††305))6*;4826)4†.)4†);806*;48†8¶(60))85;;]8*;;†*8†83"
```

```
    "(88)5*†;46(;88*96*?;8)*†(;485);5*†2:*†(;4956*2(5*—4)8¶8*"
```

```
    ";4069285);)6†8)4††;1(†9;48081;8:8†1;48†85;4)485†528806*81"
```

```
    "(†9;48;(88;4(†?34;48)4†;161;:188;†?;";
```

```
    int freq[256] = {0};
```

```
    int i;
```

```
    for (i = 0; cipher[i] != '\0'; i++)
```

```
    {
```

```
        unsigned char ch = (unsigned char)cipher[i];
```

```
        if (ch != ' ' && ch != '\n')
```

```
            freq[ch]++;
```

```
    }
```

```
    printf("Symbol Frequency Analysis\n");
```

```
    printf("-----\n");
```

```
    for (i = 0; i < 256; i++)
```

```
    {
```

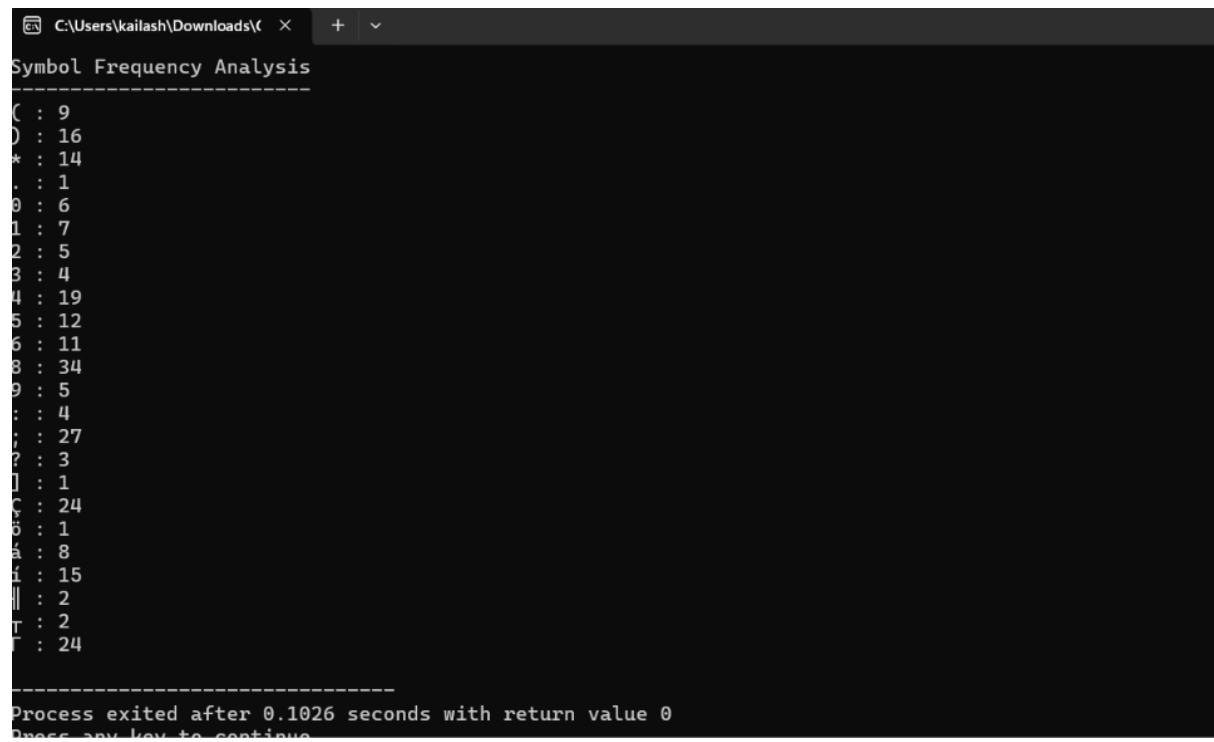
```
        if (freq[i] > 0)
```

```
            printf("%c : %d\n", i, freq[i]);
```

```
    }
```

```
    return 0;
}
```

Output:



```
C:\Users\kailash\Downloads\>
Symbol Frequency Analysis
-----
( : 9
) : 16
* : 14
. : 1
0 : 6
1 : 7
2 : 5
3 : 4
4 : 19
5 : 12
6 : 11
8 : 34
9 : 5
: : 4
; : 27
? : 3
[] : 1
ç : 24
ö : 1
ä : 8
í : 15
|| : 2
T : 2
r : 24
-----
Process exited after 0.1026 seconds with return value 0
Press any key to continue
```